

Mathematics of Cryptography

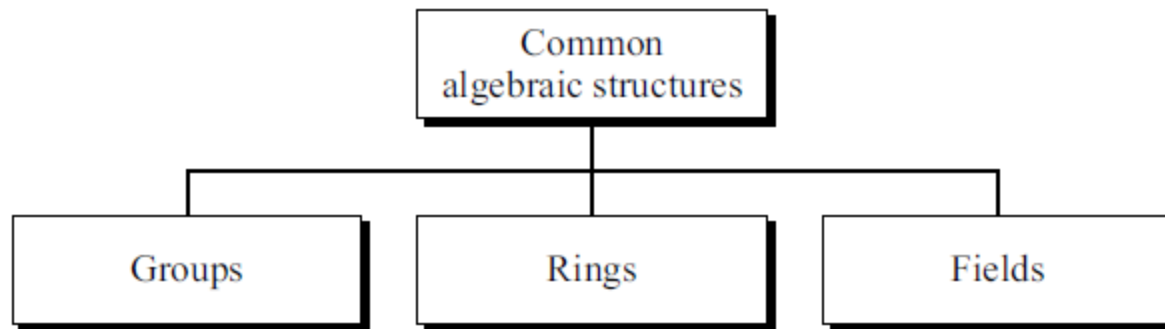
Part II

Algebraic Structures

Objectives

- To review the concept of algebraic structures
- To define and give some examples of groups
- To define and give some examples of rings
- To define and give some examples of fields
- To emphasize the finite fields of type $GF(2^n)$ that make it possible to perform operations such as addition, subtraction, multiplication, and division on n-bit words in modern block ciphers

- Groups, rings, and fields are the fundamental elements of a branch of mathematics known as algebraic structure.



Groups

Groups

- A group (G) is a set of elements with a binary operation “ $\bullet$ ” that satisfies four properties (axioms).
- Sometimes denoted by $\{G, \bullet\}$.

- Axioms

- **(A1) Closure:** If a and b belong to G , then $a \bullet b$ is also in G .
- **(A2) Associative:** $a \bullet (b \bullet c) = (a \bullet b) \bullet c$ for all a, b, c in G .
- **(A3) Identity element:** There is an element e in G such that $a \bullet e = e \bullet a = a$ for all a in G
- **(A4) Inverse element:** For each a in G , there is an element a' in G such that $a \bullet a' = a' \bullet a = e$.

Abelian group

- It is a group in which the operator satisfies the four properties for groups plus an extra property, **commutativity**.
- **(A5) Commutative:**
$$a \bullet b = b \bullet a \text{ for all } a, b \text{ in } G.$$

Application of Group

- Although a group involves a single operation, the properties imposed on the operation allow the use of a pair of operations as long as they are inverses of each other.
 - if the defined operation is addition, the group supports both addition and subtraction
 - This is also true for multiplication and division.

- When the group operation is addition, the identity element is 0
- The inverse element of a is $-a$
- subtraction is defined with the following rule:

$$a - b = a + (-b).$$

Example 1

$G = \langle \mathbb{Z}_n, + \rangle$, is a commutative group

- Closure is satisfied. The result of adding two integers in $\mathbb{Z}_n$ is another integer in $\mathbb{Z}_n$.
- Associativity is satisfied. The result of $4 + (3 + 2)$ is the same as $(4 + 3) + 2$.
- Commutativity is satisfied. We have $3 + 5 = 5 + 3$.
- The identity element is 0. We have $3 + 0 = 0 + 3 = 3$.
- Every element has an additive inverse. The inverse of an element is its complement

Example 2

The set $\mathbb{Z}_n^*$ with the multiplication operator, $G = \langle \mathbb{Z}_n^*, \times \rangle$, is also an abelian group

- Closure is satisfied.
- Associativity is satisfied.
- Commutativity is satisfied.
- The identity element is 1
- Each element has an inverse.

Finite Group

- A group is called a finite group if the set has a finite number of elements
- otherwise, it is an infinite group.

Order of a Group

- The order of a group, $|G|$, is the number of elements in the group.
- If the group is not finite, its order is infinite
- If the group is finite, the order is finite.

Subgroups

- A subset H of a group G is a subgroup of G if H itself is a group with respect to the operation on G .
- if $G = \langle S, \bullet \rangle$ is a group, $H = \langle T, \bullet \rangle$ is a group under the same operation, and T is a nonempty subset of S , then H is a subgroup of G

Subgroups

- If a and b are members of both groups, then $c = a \bullet b$ is also a member of both groups.
- The groups share the same identity element.
- If a is a member of both groups, the inverse of a is also a member of both groups.
- The group made of the identity element of G , $H = \langle \{e\}, \bullet \rangle$, is a subgroup of G .
- Each group is a subgroup of itself.

Example

Is the group $H = \langle \mathbb{Z}_{10}, + \rangle$ a subgroup of the group $G = \langle \mathbb{Z}_{12}, + \rangle$?

- No.
- Although H is a subset of G , the operations defined for these two groups are different

Cyclic Subgroups

- If a subgroup of a group can be generated using the power of an element, the subgroup is called the cyclic subgroup.
- The term power here means repeatedly applying the group operation to the element
$$a^n \rightarrow a \bullet a \bullet \dots \bullet a \text{ (n times)}$$
- The set made from this process is referred to as $\langle a \rangle$.
- when the operation is addition, a^n means multiplying n by a .

Example

- Four cyclic subgroups can be made from the group $G = \langle \mathbb{Z}_6, + \rangle$.
- They are
 - $H_1 = \langle \{0\}, + \rangle$
 - $H_2 = \langle \{0, 2, 4\}, + \rangle$
 - $H_3 = \langle \{0, 3\}, + \rangle$
 - $H_4 = G$

- The cyclic subgroup generated from 0 is H_1 , which has only one element, the identity element.

$$0^0 \bmod 6 = 0$$

- The cyclic subgroup generated from 1 is H_4 , which is G itself

$$1^0 \bmod 6 = 0$$

$$1^1 \bmod 6 = 1$$

$$1^2 \bmod 6 = (1 + 1) \bmod 6 = 2$$

$$1^3 \bmod 6 = (1 + 1 + 1) \bmod 6 = 3$$

$$1^4 \bmod 6 = (1 + 1 + 1 + 1) \bmod 6 = 4$$

$$1^5 \bmod 6 = (1 + 1 + 1 + 1 + 1) \bmod 6 = 5$$

- The cyclic subgroup generated from 2 is H_2 , which has three elements: 0, 2, and 4.

$$\begin{aligned}2^0 \bmod 6 &= 0 \\2^1 \bmod 6 &= 2 \\2^2 \bmod 6 &= (2 + 2) \bmod 6 = 4\end{aligned}$$

- The cyclic subgroup generated from 3 is H_3 , which has two elements: 0 and 3.

$$\begin{aligned}3^0 \bmod 6 &= 0 \\3^1 \bmod 6 &= 3\end{aligned}$$

- The cyclic subgroup generated from 4 is H_2 ; this is not a new subgroup.

$$\begin{aligned}4^0 \bmod 6 &= 0 \\4^1 \bmod 6 &= 4 \\4^2 \bmod 6 &= (4 + 4) \bmod 6 = 2\end{aligned}$$

- The cyclic subgroup generated from 5 is H_4 , which is G itself.

$$\begin{aligned}5^0 \bmod 6 &= 0 \\5^1 \bmod 6 &= 5 \\5^2 \bmod 6 &= 4 \\5^3 \bmod 6 &= 3 \\5^4 \bmod 6 &= 2 \\5^5 \bmod 6 &= 1\end{aligned}$$

Cyclic Groups

- A cyclic group is a group that is its own cyclic subgroup
- A group G is *cyclic if every element of G is a power g^k (k is an integer) of a fixed element $g \in G$.*
- The element g is said to *generate the group G* or to be a *generator* of G .
- A cyclic group is always abelian and may be finite or infinite.

- The element that generates the cyclic subgroup can also generate the group itself
- If g is a generator, the elements in a finite cyclic group can be written as

$$\{e, g, g^2, \dots, g^{n-1}\}, \text{ where } g^n = e$$

- A cyclic group can have many generators

Example

- The group $G = \langle \mathbb{Z}_6, + \rangle$ is a cyclic group with two generators, $g = 1$ and $g = 5$.

Lagrange's Theorem

- Lagrange's theorem relates the order of a group to the order of its subgroup.
- Assume that G is a group, and H is a subgroup of G . If the order of G and H are $|G|$ and $|H|$, respectively, then, based on this theorem, $|H|$ divides $|G|$.

- Given a group G of order $|G|$, the orders of the potential subgroups can be easily determined if the divisors of $|G|$ can be found.
- For example, the order of the group $G = \langle \mathbb{Z}_{17}, + \rangle$ is 17.
 - The only divisors of 17 are 1 and 17.
 - This means that this group can have only two subgroups, H_1 with the identity element and $H_2 = G$.

Order of an Element

- The order of an element a in a group, $\text{ord}(a)$, is the smallest integer n such that $a^n = e$.
- The order of an element is also the order of the cyclic group it generates.

Example

- In the group $G = \langle \mathbb{Z}_6, + \rangle$, the orders of the elements are: $\text{ord}(0) = 1$, $\text{ord}(1) = 6$, $\text{ord}(2) = 3$, $\text{ord}(3) = 2$, $\text{ord}(4) = 3$, $\text{ord}(5) = 6$.

Ring

Ring

- A ring R , denoted by $\{R, +, \times\}$, is a set of elements with two binary operations, called addition and multiplication.
- The first operation must satisfy all five properties required for an abelian group.
- The second operation must satisfy only the first two. In addition, the second operation must be distributed over the first.

- **(A1–A5)** R is an abelian group with respect to addition.
- **(M1)** **Closure under multiplication:**
 - If a and b belong to R , then ab is also in R .
- **(M2)** **Associativity of multiplication:**
 - $a(bc) = (ab)c$ for all a, b, c in R .
- **(M3)** **Distributive laws:**
 - $a(b + c) = ab + ac$ for all a, b, c in R .
 - $(a + b)c = ac + bc$ for all a, b, c in R .

- A **commutative ring** is a ring in which the commutative property is also satisfied for the second the operation.
- A ring is said to be **commutative** if it satisfies the following additional condition:

(M4)

Commutativity of multiplication:

$$ab = ba \text{ for all } a, b \text{ in } R.$$

Example

- $R = \{\mathbb{Z}, +, \times\}$ is a commutative ring

- An **integral domain** is a commutative ring that obeys the following axioms.
- **(M5)** Multiplicative identity:
 - There is an element 1 in R such that,
 $a1 = 1a = a$ for all a in R.
- **(M6)** No zero divisors:
 - *If a, b in R and $ab = 0$, then either $a = 0$ or $b = 0$.*

Field

Field

- A field is a set of elements in which we can do addition, subtraction, multiplication, and division without leaving the set.
- Division is defined with the following rule:

$$a/b = a(b^{-1})$$

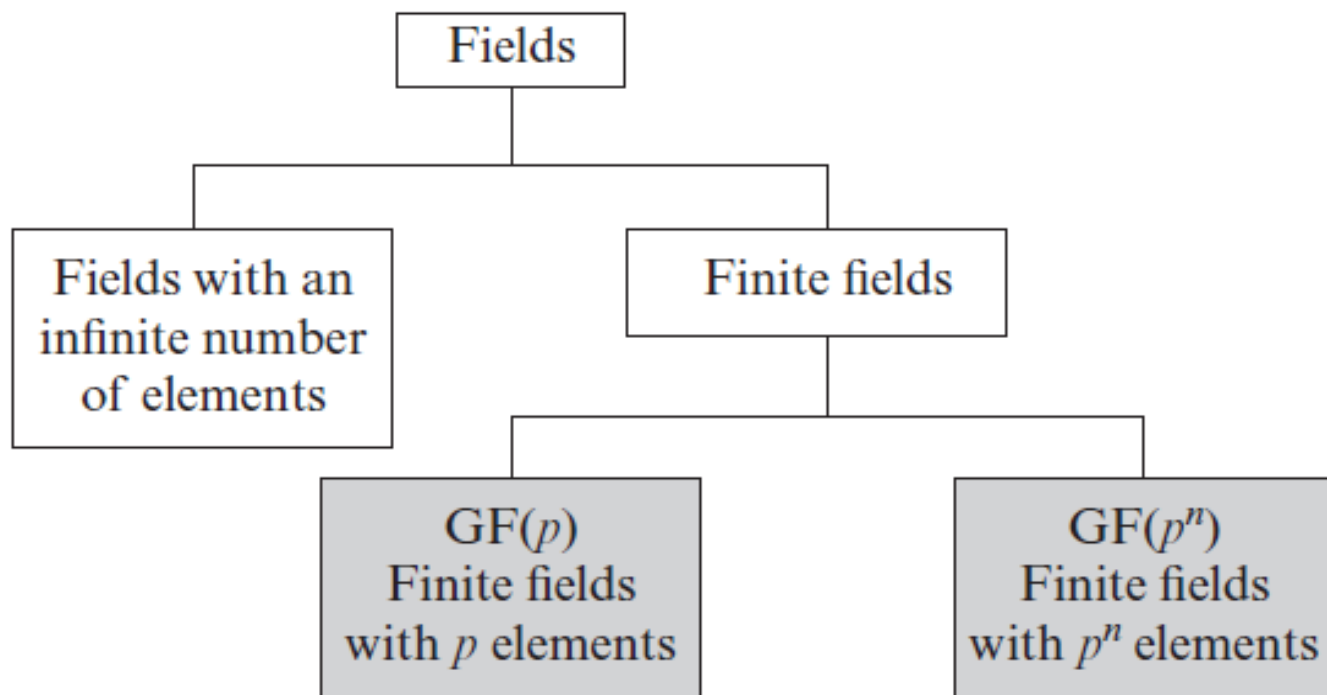
Field

- A field F is denoted by $\{F, +, \times\}$, is a set of elements with two binary operations, called addition and multiplication, such that for all a, b, c in F , the following axioms are obeyed.
- **(A1–M6)** F is an **integral domain**; that is, F satisfies axioms A1 through A5 and M1 through M6.
- **(M7)** **Multiplicative inverse**: For each a in F , except 0, there is an element a^{-1} in F such that $aa^{-1} = (a^{-1})a = 1$.

Finite Fields

- A finite field, a field with a finite number of elements.
- The finite fields are usually called **Galois fields** and denoted as $GF(p^n)$, where p is a prime and n is a positive integer.

Types of Fields



Finite Fields of Order p

- When $n = 1$, we have Finite Fields of Order p or $\text{GF}(p)$ field.
- This field can be the set $\mathbb{Z}_p$, $\{0, 1, \dots, p - 1\}$, with two arithmetic operations (addition and multiplication).
 - Each element has an additive inverse and every nonzero element have a multiplicative inverse

Example 1: GF(2) field

- The set has only 2 elements, 0 and 1
- Addition/subtraction in GF(2) is the same as the XOR operation.
- multiplication/division is the same as the AND operation.

GF(2)



+	0	1
0	0	1
1	1	0

Addition

$\cdot$	0	1
0	0	0
1	0	1

Multiplication

a	0	1
$-a$	1	0

a	0	1
a^{-1}	—	1

Inverses

Example 2: GF(5) field

- We can define GF(5) on the set Z_5 (5 is a prime) with addition and multiplication operators

GF(5)

$\{0, 1, 2, 3, 4\}$ $+$ $\times$

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Addition

$\times$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Multiplication

Additive inverse

a	0	1	2	3	4
$-a$	0	4	3	2	1

a	0	1	2	3	4
a^{-1}	—	1	3	2	4

Multiplicative inverse

- It is easy to find the multiplicative inverse of an element in $GF(p)$ for small values of p (using multiplication table).
- For large values of p , extended Euclidean algorithm is used to find multiplicative inverses.

$GF(p^n)$ Field

- The $GF(2^n)$ is a very useful field in cryptography.
- The positive integers are stored in the computer as n -bit words in which n is usually 8, 16, 32, 64, and so on.
- The range of integers is 0 to $2^n - 1$.
- The modulus is 2^n

- We have two choices if we want to use a field:
 - We can use $GF(p)$ with the set Z_p , where p is the largest prime number less than 2^n .
 - Drawback of this is we cannot use the integers from p to $2^n - 1$.
 - We can work in $GF(2^n)$ and uses a set of 2^n elements.
 - The elements in this set are n -bit words. For example, if $n = 3$, the set is
 $\{000, 001, 010, 011, 100, 101, 110, 111\}$

- There is a need to define a set of n bit words and two operations satisfying the properties defined for a field.
- The finite field $GF(2^n)$ is useful in this scenario.

GF (2^2)

- The set has four 2 bit words {00,01,10,11}
- Addition and multiplication for this field can be defined in such a way that all properties of a field are satisfied.

An example of a $GF(2^2)$ field

Addition					Multiplication				
$\oplus$	00	01	10	11	$\otimes$	00	01	10	11
00	00	01	10	11	00	00	00	00	00
01	01	00	11	10	01	00	01	10	11
10	10	11	00	01	10	00	10	11	01
11	11	10	01	00	11	00	11	01	10
Identity: 00					Identity: 01				

- Each word is the additive inverse of itself.
- Every word (except 00) has a multiplicative inverse.
- Addition and multiplication are defined in terms of **polynomials**.

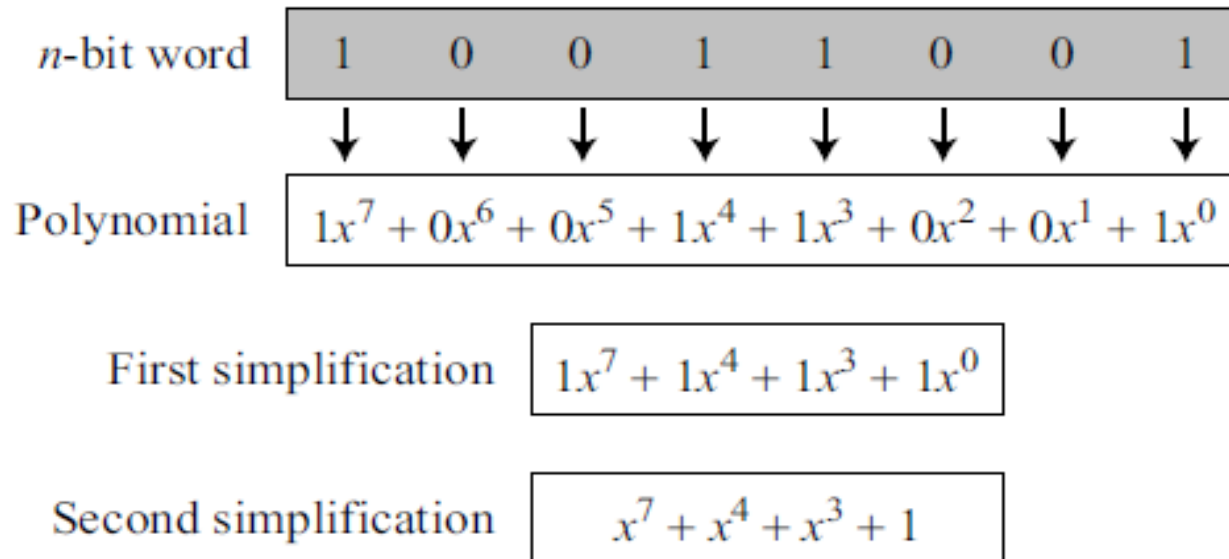
Polynomials

- A polynomial of degree $n - 1$ is an expression of the form

$$f(x) = a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x^1 + a_0x^0$$

- The power of x defines the position of the bit in the n -bit word.
- The polynomial coefficients can be either 0 or 1

- Representation of an 8-bit word by a polynomial



- To find the 8-bit word related to the polynomial $x^5 + x^2 + x$, the expanded polynomial is:

$$0x^7 + 0x^6 + 1x^5 + 0x^4 + 0x^3 + 1x^2 + 1x^1 + 0x^0$$

The 8-bit word is 00100110

Operations on Polynomials

- Any operation on polynomials actually involves two operations:
 - operations on coefficients
 - operations on two polynomials
- Polynomials representing n-bit words use two fields: $GF(2)$ and $GF(2^n)$.

Modulus of Polynomials

- Addition of two polynomials never creates a polynomial out of the set.
- Multiplication of two polynomials may create a polynomial with degrees more than $n - 1$.
- For the sets of polynomials in $GF(2^n)$, a group of polynomials of degree n is defined as the modulus.

Modulus of Polynomials

- The modulus in this case acts as a **prime polynomial** (no polynomials in the set can divide this polynomial).
 - Also referred to as **irreducible polynomials**
- When the $GF(2^n)$ is defined, we need to declare which irreducible polynomial we are using as the modulus.

- List of irreducible polynomials:

<i>Degree</i>	<i>Irreducible Polynomials</i>
1	$(x + 1), (x)$
2	$(x^2 + x + 1)$
3	$(x^3 + x^2 + 1), (x^3 + x + 1)$
4	$(x^4 + x^3 + x^2 + x + 1), (x^4 + x^3 + 1), (x^4 + x + 1)$
5	$(x^5 + x^2 + 1), (x^5 + x^3 + x^2 + x + 1), (x^5 + x^4 + x^3 + x + 1),$ $(x^5 + x^4 + x^3 + x^2 + 1), (x^5 + x^4 + x^2 + x + 1)$

Addition of polynomials in GF(2)

- The coefficients of the corresponding terms are added in GF(2).
- Adding two polynomials of degree $n - 1$ always create a polynomial with degree $n - 1$
- No need to reduce the result using the modulus.

Example

- Add $(x^5 + x^2 + x)$ and $(x^3 + x^2 + 1)$ in $GF(2^8)$.

$$\begin{array}{rcl} 0x^7 + 0x^6 + 1x^5 + 0x^4 + 0x^3 + 1x^2 + 1x^1 + 0x^0 & \oplus & \\ 0x^7 + 0x^6 + 0x^5 + 0x^4 + 1x^3 + 1x^2 + 0x^1 + 1x^0 & & \\ \hline 0x^7 + 0x^6 + 1x^5 + 0x^4 + 1x^3 + 0x^2 + 1x^1 + 1x^0 & \rightarrow & x^5 + x^3 + x + 1 \end{array}$$

keep the uncommon terms and delete the common terms

Additive Identity

- The additive identity in a polynomial is a **zero polynomial**.
 - a polynomial with all coefficients set to zero

Additive Inverse

- The additive inverse of a polynomial with coefficients in $GF(2)$ is the polynomial itself.
- Addition and subtraction operations on polynomials are the same operation.

Multiplication in polynomials

- The coefficient multiplication is done in GF(2).
- Multiplying x^i by x^j results in x^{i+j}
- Multiplication may create terms with degree more than $n - 1$, which means the result needs to be reduced using a modulus polynomial

Example

- Multiply $(x^5 + x^2 + x)$ with $(x^7 + x^4 + x^3 + x^2 + x)$ in $GF(2^8)$ with irreducible polynomial $(x^8 + x^4 + x^3 + x + 1)$.

$$(x^5 + x^2 + x)(x^7 + x^4 + x^3 + x^2 + x)$$

$$= x^5(x^7 + x^4 + x^3 + x^2 + x) + x^2(x^7 + x^4 + x^3 + x^2 + x) + x(x^7 + x^4 + x^3 + x^2 + x)$$

$$= x^{12} + x^9 + x^8 + x^7 + x^6 + x^9 + x^6 + x^5 + x^4 + x^3 + x^8 + x^5 + x^4 + x^3 + x^2$$

$$= x^{12} + x^7 + x^2$$

$$= (x^{12} + x^7 + x^2) \bmod (x^8 + x^4 + x^3 + x + 1)$$

$$x^4 + 1$$

$$\begin{array}{r}
 x^8 + x^4 + x^3 + x + 1 \overline{) \begin{array}{l} x^{12} + x^7 + x^2 \\ x^{12} + x^8 + x^7 + x^5 + x^4 \\ \hline x^8 + x^5 + x^4 + x^2 \\ x^8 + x^4 + x^3 + x + 1 \\ \hline \end{array} } \\
 \text{Remainder} \quad \boxed{x^5 + x^3 + x^2 + x + 1}
 \end{array}$$

- **Multiplicative Identity:**
 - The multiplicative identity is always 1.
- **Multiplicative Inverse:**
 - The extended Euclidean algorithm must be applied to the modulus and the polynomial.

Example

- In $GF(2^4)$, find the inverse of $(x^2 + 1)$ modulo $(x^4 + x + 1)$.

q	r_1	r_2	r	t_1	t_2	t
$(x^2 + 1)$	$(x^4 + x + 1)$	$(x^2 + 1)$	(x)	(0)	(1)	$(x^2 + 1)$
(x)	$(x^2 + 1)$	(x)	(1)	(1)	$(x^2 + 1)$	$(x^3 + x + 1)$
(x)	(x)	(1)	(0)	$(x^2 + 1)$	$(x^3 + x + 1)$	(0)
	(1)	(0)		$(x^3 + x + 1)$	(0)	

$$[(x^2 + 1) \otimes (x^3 + x + 1)] \bmod (x^4 + x + 1) = 1$$

Multiplication Using Computer

- The computer implementation uses a better algorithm, repeatedly multiplying a reduced polynomial by x .
- For example, instead of finding the result of $(x^2 \otimes P2)$, the program finds the result of $(x \otimes (x \otimes P2))$.

- Find the result of multiplying $P_1 = (x^5 + x^2 + x)$ by $P_2 = (x^7 + x^4 + x^3 + x^3 + x)$ in $GF(2^8)$ with irreducible polynomial $(x^8 + x^4 + x^3 + x + 1)$

<i>Powers</i>	<i>Operation</i>	<i>New Result</i>	<i>Reduction</i>
$x^0 \otimes P_2$		$x^7 + x^4 + x^3 + x^2 + x$	No
$x^1 \otimes P_2$	$x \otimes (x^7 + x^4 + x^3 + x^2 + x)$	$x^5 + x^2 + x + 1$	Yes
$x^2 \otimes P_2$	$x \otimes (x^5 + x^2 + x + 1)$	$x^6 + x^3 + x^2 + x$	No
$x^3 \otimes P_2$	$x \otimes (x^6 + x^3 + x^2 + x)$	$x^7 + x^4 + x^3 + x^2$	No
$x^4 \otimes P_2$	$x \otimes (x^7 + x^4 + x^3 + x^2)$	$x^5 + x + 1$	Yes
$x^5 \otimes P_2$	$x \otimes (x^5 + x + 1)$	$x^6 + x^2 + x$	No
$P_1 \times P_2 = (x^6 + x^2 + x) + (x^6 + x^3 + x^2 + x) + (x^5 + x^2 + x + 1) = x^5 + x^3 + x^2 + x + 1$			

- Multiplication of polynomials in $GF(2^n)$ can be achieved using **shift-left** and **exclusive-OR** operations.
- A simple algorithm can be constructed:
 1. If the most significant bit of the previous result is 0, just shift the previous result one bit to the left.
 2. If the most significant bit of the previous result is 1, then,
 - a. shift it one bit to the left, and
 - b. exclusive-or it with the modulus without the most significant bit

- Suppose $P_1 = 000100110$, $P_2 = 10011110$, and modulus = 100011010

<i>Powers</i>	<i>Shift-Left Operation</i>	<i>Exclusive-Or</i>
$x^0 \otimes P_2$		10011110
$x^1 \otimes P_2$	00111100	$(00111100) \oplus (00011010) = \underline{00100111}$
$x^2 \otimes P_2$	01001110	<u>01001110</u>
$x^3 \otimes P_2$	10011100	10011100
$x^4 \otimes P_2$	00111000	$(00111000) \oplus (00011010) = 00100011$
$x^5 \otimes P_2$	01000110	<u>01000110</u>
$P_1 \otimes P_2 = (00100111) \oplus (01001110) \oplus (01000110) = 00101111$		

Operation of Polynomials using generators

- It is easier to define the elements of the $GF(2^n)$ field using a generator.
- In this field with the irreducible polynomial $f(x)$, an element in the field, a , must satisfy the relation $f(a) = 0$.
- If g is a generator of the field, then $f(g) = 0$.
- The elements of the field can be generated as $\{0, g^0, g^1, g^2, \dots, g^N\}$, where $N = 2^n - 2$

Example

- Generate the elements of the field $GF(2^4)$ using the irreducible polynomial $f(x) = x^4 + x + 1$.
- The relation can be written as,

$$f(g) = g^4 + g + 1 = 0$$

$$g^4 = -g - 1 \quad \text{or} \quad g^4 = g + 1$$

As in this field, addition and subtraction are same.

Here, $N=2^n - 2 = 14$

So, the elements are $\{0, g^0, g^1, \dots, g^{14}\}$

$$0 \rightarrow 0000$$

$$g^0 \rightarrow 0001$$

$$g^1 \rightarrow 0010$$

$$g^2 \rightarrow 0100$$

$$g^3 \rightarrow 1000$$

$$g^4 \rightarrow g + 1 \rightarrow 0011$$

$$g^5 \rightarrow g(g^4) \rightarrow g(g + 1) \rightarrow g^2 + g \rightarrow 0100 + 0010 \rightarrow 0110$$

$$g^6 \rightarrow g(g^5) \rightarrow g(g^2 + g) \rightarrow g^3 + g^2 \rightarrow 1000 + 0100 \rightarrow 1100$$

$$g^7 \rightarrow g(g^6) \rightarrow g(g^3 + g^2) \rightarrow g^4 + g^3 \rightarrow g^3 + g + 1 \rightarrow 1000 + 0010 + 1 \rightarrow 1011$$

$$g^8 \rightarrow g(g^7) \rightarrow g(g^3 + g + 1) \rightarrow g^4 + g^2 + g \rightarrow g + 1 + g^2 + g \rightarrow g^2 + 1 \rightarrow 0100 + 1 \rightarrow 0101$$

$$g^9 \rightarrow g(g^8) \rightarrow g(g^2 + 1) \rightarrow g^3 + g \rightarrow 1000 + 0010 \rightarrow 1010$$

$$g^{10} \rightarrow g(g^9) \rightarrow g(g^3 + g) \rightarrow g^4 + g^2 \rightarrow g + 1 + g^2 \rightarrow 0010 + 1 + 0100 \rightarrow 0111$$

$$g^{11} \rightarrow g(g^{10}) \rightarrow g(g + 1 + g^2) \rightarrow g^3 + g^2 + g \rightarrow 1000 + 0100 + 0010 \rightarrow 1110$$

$$g^{12} \rightarrow g(g^{11}) \rightarrow g(g^3 + g^2 + g) \rightarrow g^4 + g^3 + g^2 \rightarrow g + 1 + g^3 + g^2 \rightarrow 1000 + 0100 + 1 \rightarrow 1101$$

$$g^{13} \rightarrow g(g^{12}) \rightarrow g(g + 1 + g^3 + g^2) \rightarrow g^3 + g^2 + 1 \rightarrow 1000 + 0100 + 1 \rightarrow 1101$$

$$g^{14} \rightarrow g(g^{13}) \rightarrow g(g^3 + g^2 + 1) \rightarrow g^4 + g^3 + g \rightarrow g + 1 + g^3 + g \rightarrow g^3 + 1 \rightarrow 1000 + 1 \rightarrow 1001$$

Inverses

- Additive Inverses

- The additive inverse of each element is the element itself.

- Multiplicative Inverses

- The exponents are calculated modulo $2^n - 1$

$$(g^3)^{-1} = g^{-3} = g^{12} = g^3 + g^2 + g + 1 \rightarrow (1111)$$

g^3 and g^{12} are inverses of each other because,

$$g^3 \times g^{12} = g^{15} = g^0 = 1$$

Operations

- Addition and Subtraction
 - Addition and subtraction are the same operation
 - For example,
$$g^3 - g^6 = g^3 + g^6 = g^3 + (g^3 + g^2) = g^2 \rightarrow (0100)$$

Operations

- Multiplication and Division
 - The Multiplication is the addition of powers modulo $2^n - 1$.
 - Division is multiplication using the multiplicative inverse.

For example,

$$g^9 \times g^{11} = g^{20} = g^{20 \bmod 15} = g^5 = g^2 + g \rightarrow (0110)$$

$$g^3 / g^8 = g^8 \times g^7 = g^{10} = g^2 + g + 1 \rightarrow (0111)$$